

Learning Objectives:

- Ability to identify the characteristics of programming paradigms and phases of translation
- Ability to understand the concepts of formal syntax and semantics of programming languages
- Ability to understand the concept of language-based security
- Ability to understand the concept of web scripting security
- Ability to understand the concepts of first-class values, lists, and recursion
- Ability to understand the concepts of the functional programming paradigm and design functional programs
- Ability to understand the concepts of the logic programming paradigm and design logic programs

Tentative Course Schedule:

Week	Date	Material Covered
1	Aug 26, Aug 28	Syllabus, characteristics of programming paradigms (Sebesta-Ch1), Functional programming paradigm (Sebesta- Ch15)
2	Sep 2, Sep 4	Labor Day- Functional programming paradigm (Sebesta- Ch15)
3	Sep 9, Sep 11	Functional programming paradigm (Sebesta-Ch15)
4	Sep 16, Sep 18	Describing syntax and semantic (Sebesta-Ch3)
5	Sep 23, Sep 25	Describing syntax and semantic (Sebesta-Ch3)
6	Sep 30, Oct 2	Lexical and syntax analysis (Sebesta-Ch4)
7	Oct 7, Oct 9	Lexical and syntax analysis (Sebesta-Ch4)
8	Oct 14, Oct 16	Common Weakness Enumeration (CWE), Midterm Exam
9	Oct 21, Oct 23	Logical programming paradigm (Sebesta-Ch16)
10	Oct 28, Oct 30	Logical programming paradigm (Sebesta-Ch16)
11	Nov 4, Nov 6	Language based security (layered security) and Web Scripting Security
12	Nov 11, Nov 13	Language based security (layered security) and Web Scripting Security , Language based security (type-safe vs. type-unsafe languages)
13	Nov 18, Nov 20	Language based security (Formal method-Fuzzing)
14	Nov 25, Nov 27	Fall Break
15	Dec 2, Dec 4	Language based security (Fuzzing, control-flow integrity)
16	Dec 9	Final Exam